

Logistic Regression **Classifiers**

What is logistic regression?

- This type of statistical model (also known as *logit model*) is often used for classification and predictive analytics.
- Logistic regression estimates the probability of an event occurring, such as voted or didn't vote, based on a given dataset of independent variables.
- Since the outcome is a probability, the dependent variable is bounded between 0 and 1.
- In logistic regression, a logit transformation is applied on the odds—that is, the probability of success divided by the probability of failure. This is also commonly known as the log odds, or the natural logarithm of odds, and this logistic function is represented by the following formulas:
- Logit(pi) = 1/(1+ exp(-pi))
- ln(pi/(1-pi)) = Beta_0 + Beta_1*X_1 + … + B_k*K_k

Ensemble Learning

- ln(pi/(1-pi)) = Beta_0 + Beta_1*X_1 + … + B_k*K_k
- In this logistic regression equation ln is the dependent or response variable and x is the independent variable.
- The beta parameter, or coefficient, in this model is commonly estimated via maximum likelihood estimation (MLE). This method tests different values of beta through multiple iterations to optimize for the best fit of log odds.
- In machine learning Gradient Descend is used instead of MLE.
- For binary classification, a probability less than .5 will predict 0 while a probability greater than 0 will predict 1.
- After the model has been computed, it's best practice to evaluate the how well the model predicts the dependent variable, which is called goodness of fit. The Hosmer–Lemeshow (HL test) test is a popular method to assess model fit.

Interpreting logistic regression

- Log odds can be difficult to make sense of within a logistic regression data analysis. As a result, exponentiating the beta estimates is common to transform the results into an odds ratio (OR), easing the interpretation of results.
- The OR (odds ratio) represents the odds that an outcome will occur given a particular event, compared to the odds of the outcome occurring in the absence of that event.
- If the OR is greater than 1, then the event is associated with a higher odds of generating a specific outcome.
- Conversely, if the OR is less than 1, then the event is associated with a lower odds of that outcome occurring.
- Based on the equation from above, the interpretation of an odds ratio can be denoted as the following: the odds of a success changes by exp(cB_1) times for every c-unit increase in x. To use an example, let's say that we were to estimate the odds of survival on the Titanic given that the person was male, and the odds ratio for males was .0810. We'd interpret the odds ratio as the odds of survival of males decreased by a factor of .0810 when compared to females, holding all other variables constant.

Types of Logistic Regression

- There are three types of logistic regression models, which are defined based on categorical response.
- **Binary logistic regression:** In this approach, the response or dependent variable is dichotomous in nature—i.e. it has only two possible outcomes (e.g. 0 or 1). Some popular examples of its use include predicting if an e-mail is spam or not spam or if a tumor is malignant or not malignant. Within logistic regression, this is the most commonly used approach, and more generally, it is one of the most common classifiers for binary classification.
- **Multinomial logistic regression:** In this type of logistic regression model, the dependent variable has three or more possible outcomes; however, these values have no specified order. For example, movie studios want to predict what genre of film a moviegoer is likely to see to market films more effectively. A multinomial logistic regression model can help the studio to determine the strength of influence a person's age, gender, and dating status may have on the type of film that they prefer. The studio can then orient an advertising campaign of a specific movie toward a group of people likely to go see it.
- **Ordinal logistic regression:** This type of logistic regression model is leveraged when the response variable has three or more possible outcome, but in this case, these values do have a defined order. Examples of ordinal responses include grading scales from A to F or rating scales from 1 to 5.

Logistic regression and machine learning

- Within machine learning, logistic regression belongs to the family of supervised machine learning models. It is also considered a discriminative model, which means that it attempts to distinguish between classes (or categories). Unlike a generative algorithm, such as naïve bayes, it cannot, as the name implies, generate information, such as an image, of the class that it is trying to predict (e.g. a picture of a cat).
- Previously, we mentioned how logistic regression maximizes the log likelihood function to determine the beta coefficients of the model. This changes slightly under the context of machine learning. Within machine learning, the negative log likelihood used as the loss function, using the process of gradient descent to find the global maximum. This is just another way to arrive at the same estimations discussed above.
- Logistic regression can also be prone to overfitting, particularly when there is a high number of predictor variables within the model. Regularization is typically used to penalize parameters large coefficients when the model suffers from high dimensionality.
- Scikit-learn provides valuable documentation to learn more about the logistic regression machine learning model.

Categorical Response Variables

Whether or not a person smokes

Binary Response

$$Y = \begin{cases} \text{Non - smoker} \\ \text{Smoker} \end{cases}$$

Success of a medical treatment

$$Y = \begin{cases} \text{Survives} \\ \text{Dies} \end{cases}$$

Opinion poll responses

Ordinal Response

$$Y = \begin{cases} \text{Agree} \\ \text{Neutral} \\ \text{Disagree} \end{cases}$$

Example: Height predicts Gender

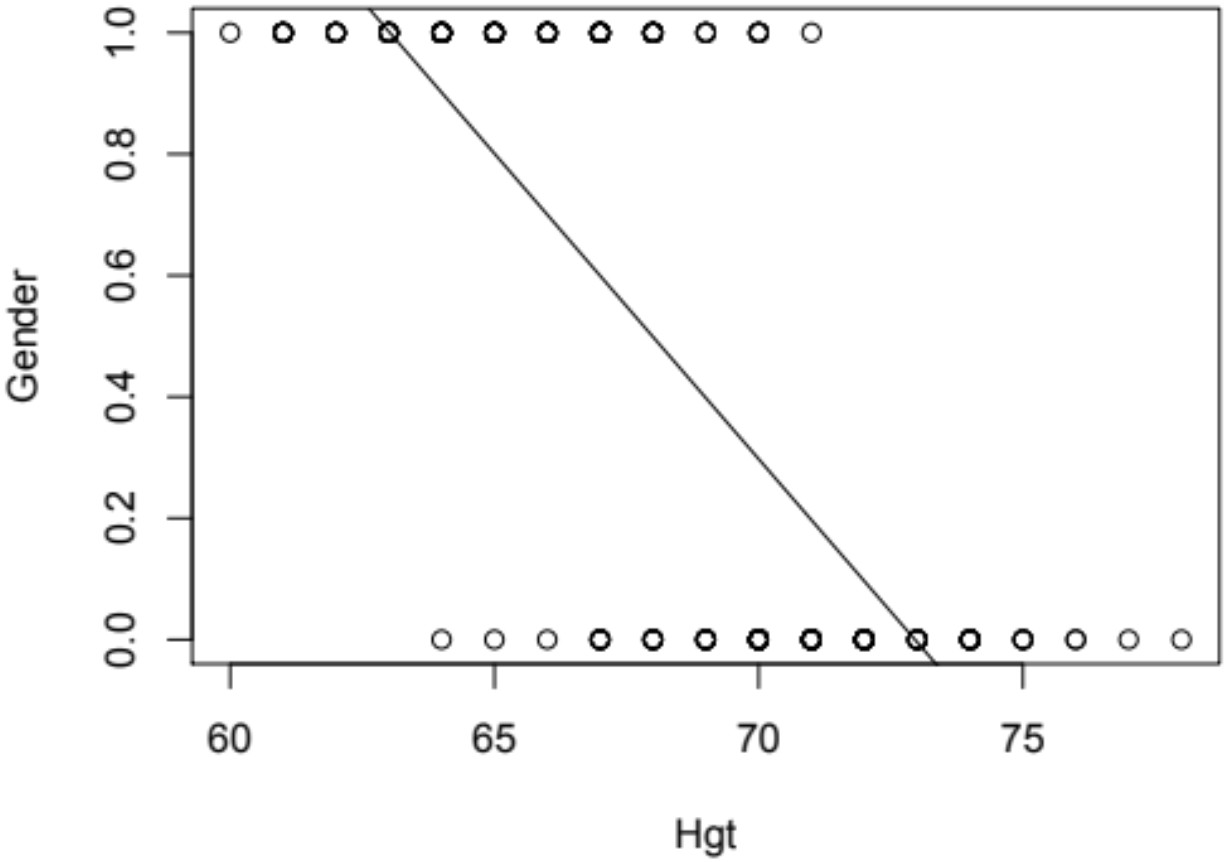
Y = Gender (0=Male 1=Female)

X = Height (inches)

Try an ordinary linear regression

```
> regmodel=lm(Gender~Hgt,data=Pulse)
> summary(regmodel)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.343647   0.397563   18.47   <2e-16 ***
Hgt         -0.100658   0.005817  -17.30   <2e-16 ***
```


Gender
0.0
0.2
0.4
0.6
0.8
1.0
60
65
70
75
Hgt

π = Proportion of “Success”

In ordinary regression the model predicts the *mean* Y for any combination of predictors.

What’s the “mean” of a 0/1 indicator variable?

$$\bar{y} = \frac{\sum y_i}{n} = \frac{\#\text{ of } 1's}{\#\text{ of trials}} = \text{Proportion of "success"}$$

Goal of logistic regression: Predict the “true” proportion of success, π, at any value of the predictor.

Binary Logistic Regression Model

Y = Binary response

X = Quantitative predictor

π = proportion of 1's (yes,success) at any X

Equivalent forms of the logistic regression model:

Logit form

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X$$

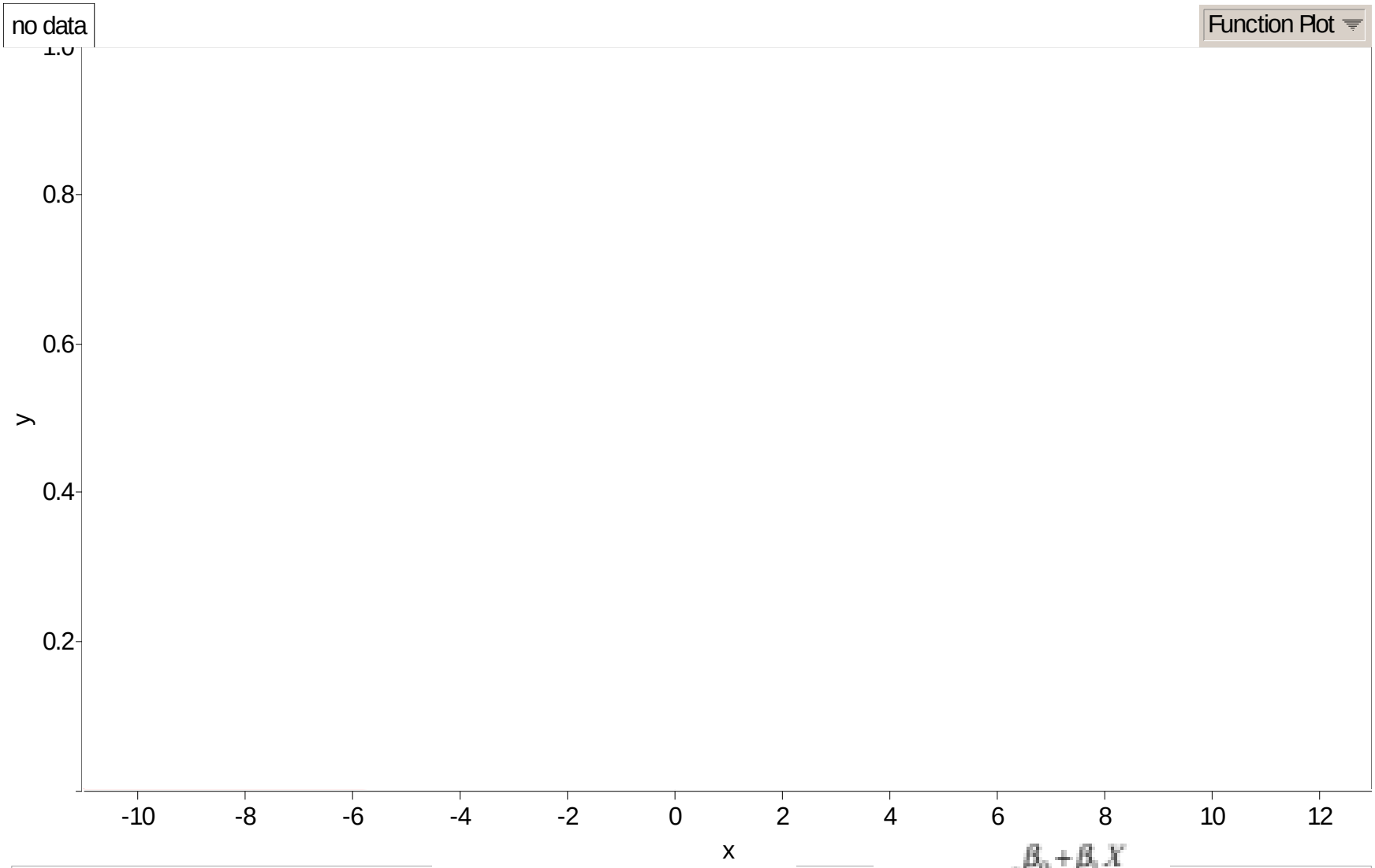
Probability form

$$\pi = \frac{e^{\beta_0+\beta_1 X}}{1+e^{\beta_0+\beta_1 X}}$$

What does this look like?

N.B.: This is natural log (aka "ln")

Logit Function



$$y = \frac{\exp(bo + b1 \cdot x)}{1 + \exp(bo + b1 \cdot x)}$$

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X$$

$$\pi = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

Binary Logistic Regression via R

```
> logitmodel=glm(Gender~Hgt,family=binomial,
data=Pulse)
> summary(logitmodel)

Call:
glm(formula = Gender ~ Hgt, family = binomial)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-2.77443  -0.34870  -0.05375   0.32973   2.37928

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  64.1416     8.3694   7.664 1.81e-14 ***
Hgt          -0.9424     0.1227  -7.680 1.60e-14***
---
```

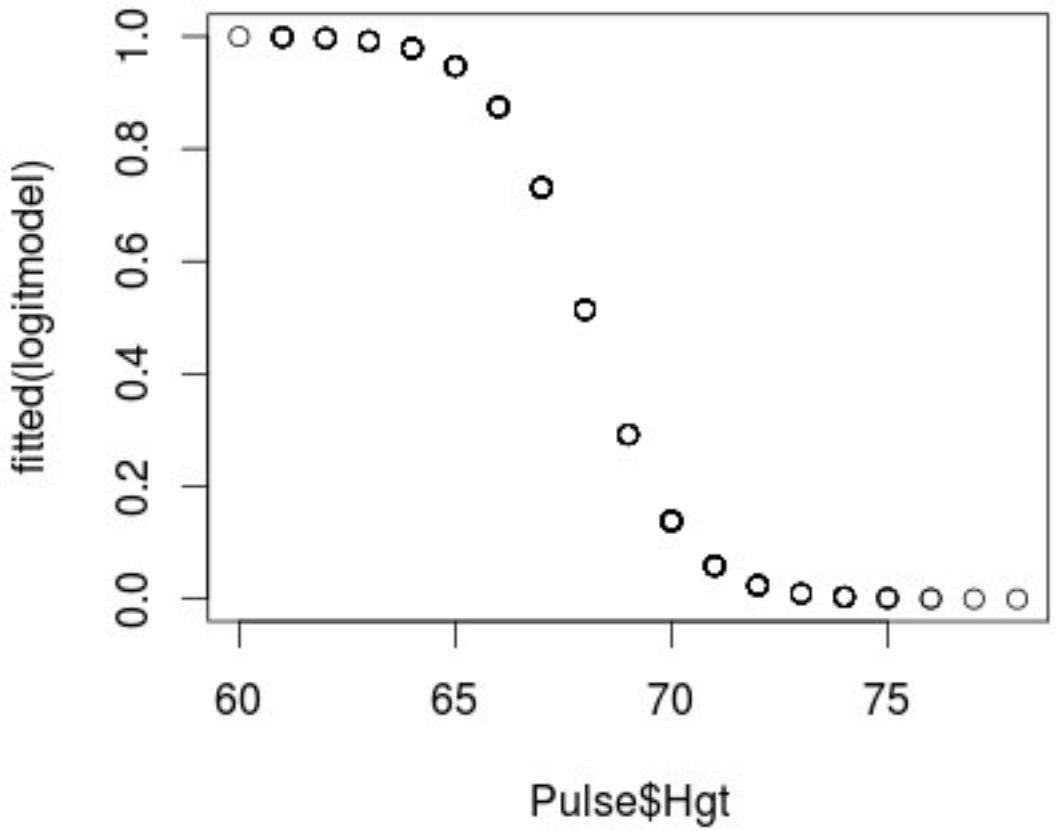
```
Call:
glm(formula = Gender ~ Hgt, family = binomial, data = Pulse)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  64.1416     8.3694   7.664 1.81e-14 ***
Hgt          -0.9424     0.1227  -7.680 1.60e-14***
---
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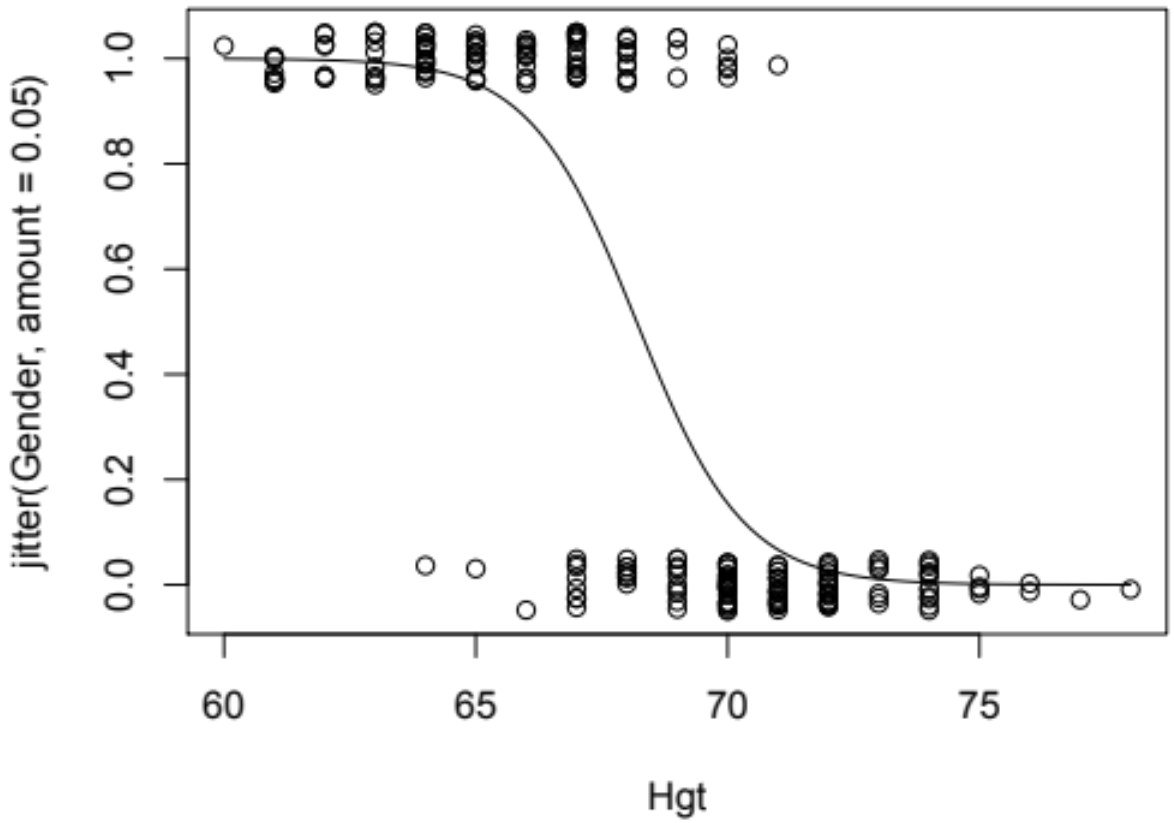
$$\hat{\pi} = \frac{e^{64.14 - 0.9424\,Hr}}{1 + e^{64.14 - .9424\,Hr}}$$

proportion of females at that Hgt

```
> plot(fitted(logitmodel)~Pulse$Hgt)
```



```
> with(Pulse,plot(Hgt,jitter(Gender,amount=0.05)))
> curve(exp(64.1-0.94*x)/(1+exp(64.1-0.94*x)), add=TRUE)
```



References

- [1] https://www.ibm.com/ae-en/topics/logistic-regression
- [2] Slides by Jeff Witmer: https://www.oberlin.edu/jeff-witmer



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Bye Bye! ☺